Assignment 03

For this assignment you will be writing 4 programs which should be saved independently as their own ".py" file. The filename you should use for each program is outlined in the sections below. For the final problem, you will be submitting a .doc or .pdf file. When you're finished you should submit your programs to the Assignment #3 category inside of Brightspace.

Problem 0. Two rectangles (Warm-Up)

- Prompt the user to enter in the dimensions of two different rectangles.
- Calculate the area of each rectangle. Area of a rectangle is computed by multiplying length by width.
- Determine which rectangle is larger and print out the result. If the rectangles are the same you should tell the user with a special print statement (i.e. Rectangle #1 and Rectangle #2 are the same size)
- Determine if either rectangle is a square. If so, tell the user.
- Comment your source code and describe your code to someone who may be viewing it for the first time.
- Here is a sample running of the program (user input is underlined):

```
Enter the width for Rectangle #1: 5

Enter the length for Rectangle #1: 10

Enter the width for Rectangle #2: 3

Enter the length for Rectangle #2: 5

Rectangle #1 has an area of 50

Rectangle #2 has an area of 15

Rectangle #1 is larger than Rectangle #2
```

- You may assume that your user will only enter integers for this program.

Problem 0 Points (18):

- We will test 3 different rectangles, and each will be worth 6 points.
    - Test 1: “Rectangle #1 and Rectangle #2 are the same size!” is printed and area of each rectangle is correct.
    - Test 2: “Rectangle #1 is larger than Rectangle #2!” is printed and area of each rectangle is correct.
    - Test 3: “Rectangle #2 is larger than Rectangle #1!” is printed and area of each rectangle is correct.
    - In at least one of these tests the rectangle will be a square. This information must also be printed.

Problem #1: What Kind of Triangle?

- Prompt the user to enter in 3 side lengths. You can assume the user will enter these values as floating point numbers.
- Next, determine if the three lengths could form a valid triangle. You can assume that the triangle is valid by checking the following.
  - Let a, b, c be the 3 sides of the triangle
  - a + b > c
  - b + c > a
  - a + c > b
- If the triangle is valid, print out the perimeter of the triangle to the nearest tenth.
- Report whether the triangle is an equilateral, isosceles, or scalene triangle:
  - Equilateral Triangle: all sides of the triangle have the same length
  - Isosceles Triangle: only two sides of the triangle have the same length
  - Scalene Triangle: all sides of the triangle have different lengths
  - Hint: when comparing the size of each triangle it may be helpful to compare a "rounded" or "formatted" version of the length of each side. Due to floating point inaccuracies you may run into a where two values are virtually identical but Python will evaluate them to be different (i.e. `0.000000001 != 0.0`)
- **OPTIONAL BONUS:** Also report if the triangle is a right triangle (i.e. it has one angle of 90 degrees). You can do this by using the Pythagorean Theorem on the sides of the triangle - $a^2 + b^2 = c^2$.
  - Hint: "c" (the hypotenuse) MUST be the largest side for the Pythagorean Theorem to work. Use an "if" statement to find the largest side and then test to see if the theorem holds.
  - Hint #2: round your numbers to 0 decimal points for the purpose of this calculation. For example, round(a^2) + round(b^2) == round(c^2)
- Here are some sample runnings of the program (user input is underlined):

Run #1

```
Triangle Tester


What is the length of side 1? 100

What is the length of side 2? 100

What is the length of side 3? 100

This is a valid triangle!

The perimeter of the triangle is 300.0

This is an equilateral triangle
```

Run #2

```
Triangle Tester


What is the length of side 1? 6
What is the length of side 2? 8
What is the length of side 3? 10
This is a valid triangle!
The perimeter of the triangle is 24.0
This is a scalene triangle
This is also a right triangle
```

Run #3

```
Triangle Tester


What is the length of side 1? 1
What is the length of side 2? 10
What is the length of side 3? 20
This is not a valid triangle.
```

Problem 1 Points (24):

- We will test 4 different triangles, and each will be worth 6 points.
    - Test 1: “This is a valid triangle!” is printed, followed by the correct perimeter as a float, followed by “This is an equilateral triangle”
    - Test 2: “This is a valid triangle!” is printed, followed by the correct perimeter as a float, followed by “This is an isosceles triangle”
    - Test 3: “This is a valid triangle!” is printed, followed by the correct perimeter as a float, followed by “This is a scalene triangle”
    - Test 4: “This is not a valid triangle.” is printed.
- At least one of these tests will be a right triangle. If you completed the bonus, “This is also a right triangle” should be printed at the end. This is worth 6 extra points.

Problem #2: Rock, Paper, Scissors

Write a program that asks the user to play “rock,” “paper,” or “scissors.” Note: the program should work regardless of capitalization of letters but if an invalid input is entered, the program should not accept it. Then, have the computer randomly select rock, paper, or scissors. The computer can only choose numbers, so you will have to do some “translating” for the computer.

Here are the rules:

- Rock beats scissors
- Paper beats rock
- Scissors beat paper
- If the same object is picked, it is a tie

Here are a few sample runnings of your program (underlined text indicates user input):

Run #1

```
Rock, Paper, Scissors

Rock, paper, or scissors? rock

The computer played paper.

You lose!
```

Run #2

```
Rock, Paper, Scissors

Rock, paper, or scissors? Paper

The computer played paper.

You tie!
```

Run #3

```
Rock, Paper, Scissors

Rock, paper, or scissors? sizor

That is not a valid option.

Please try again!
```

Problem 2 Points **(26)**:

- Code works regardless of capitalization (3 points)
- If something other than rock, paper, or scissors is played by the player, they get an error message (3 points)
- Rock, paper, or scissors is randomly chosen for the computer (5 points)
- Conditionals are correctly used for the different game play options: rock beats scissors, paper beats rock, scissors beats paper, and same object is a tie. The outcome is printed. (15 points)

Problem #3: Daily Schedule

You will be writing a program that first asks the user for a time of day. For simplicity, the user's schedule will be in military time. You can assume that the user will enter an integer (i.e. 0 = 12AM, 7 = 7AM, 13 = 1PM, 22 = 11PM) for this input. **However, you cannot assume this number will be between 0 and 23 and will need to make sure the time is valid (see sample run below).** Next, you will ask them for the day of week. Depending on whether it is a weekday (Monday-Friday) or a weekend (Saturday or Sunday), there are some differences to the schedule. You can assume the user will always enter a valid day of the week (i.e. Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday) for this input.

Here is the daily schedule you will use to tell the user what activity they should be doing at a given day/time:

- 8AM-3PM (8-15): Class
- 4PM-6PM (16-18): Free time
- 7PM-9PM (19-21):
  - Weekday: Cook and eat dinner
  - Weekend: Go out for dinner with friends
- 10PM-7AM (22-23, 0-7): Sleep

Here are a few sample runnings of the program (underlined text indicates user input):

What time is it? <u>19</u>

What day of the week is it? <u>Tuesday</u>

During this time, you cook and eat dinner

What time is it? <u>19</u>

What day of the week is it? <u>Sunday</u>

During this time, you go out for dinner with friends

What time is it? <u>6</u>

What day of the week is it? <u>Wednesday</u>

During this time, you sleep

What time is it? <u>27</u>

What day of the week is it? <u>Thursday</u>

```
You have entered an invalid time!
```

Problem 3 Points (32):

- The first user input (time) is converted to an integer (2 points)

- There will be 6 inputs, and each will be worth 5 points. Here are the following outcomes that should be printed:
  - Test #1: “During this time, you have class”
  - Test #2: “During this time, you sleep”
  - Test #3: “During this time, you have free time”
  - Test #4: “During this time, you cook and eat dinner”
  - Test #5: “During this time, you go out for dinner with friends”

    Test #6: “You have entered an invalid time!”